

Cloud Computing Technologies

DAT515 - Fall 2025

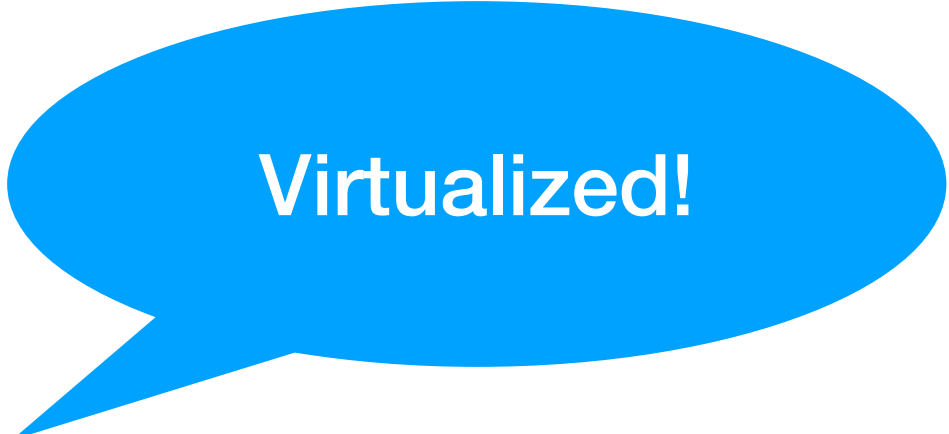
Virtualization and Containers

Prof. Hein Meling



Cloud Infrastructure

- Collection of components required to provide cloud computing
- Hardware Resources
 - Processing Units (CPU, NPU, TPU, GPU)
 - Storage Devices (RAM, SSD, HDD)
 - Network Devices (NIC, Switches, Routers)



Virtualized!



Interconnects

General-Purpose Processing Units (Servers)

Feature	Description	Typical Values
Cores	Mix of Performance (P-cores) and Efficiency (E-cores) for balancing speed vs energy	4–64 total cores in servers
SIMD/AVX	Vector instructions operating on multiple data per cycle (Single Instruction Multiple Data / Advanced Vector Extension)	128–512-bit vectors
Clock Speed	Optimized for single-thread latency	2-5 GHz
Energy Use	E-cores save power for background tasks P-cores handle bursts of heavy compute	Server CPUs: 100–300W TDP

Key Point: CPUs provide versatility, orchestrating tasks and running workloads that need low latency and strong single-thread performance.

Specialized Processing Units (Servers)

Unit	Characteristics	Typical Core Count	Typical Use
CPU	General-purpose, latency-optimized	4-64	OS, General Apps
GPU	Massively parallel, throughput-optimized (SIMD and SIMT = Thread)	1,000s	Graphics, ML training, high-performance compute
TPU	ASIC for tensor ops, systolic arrays of Multiply-Accumulate Units (for matrix/vector math)	100s-1,000s MAC Units	Deep Learning
NPU	AI-centric, low-power inference cores	10s-100s	Edge AI, Mobile Inference

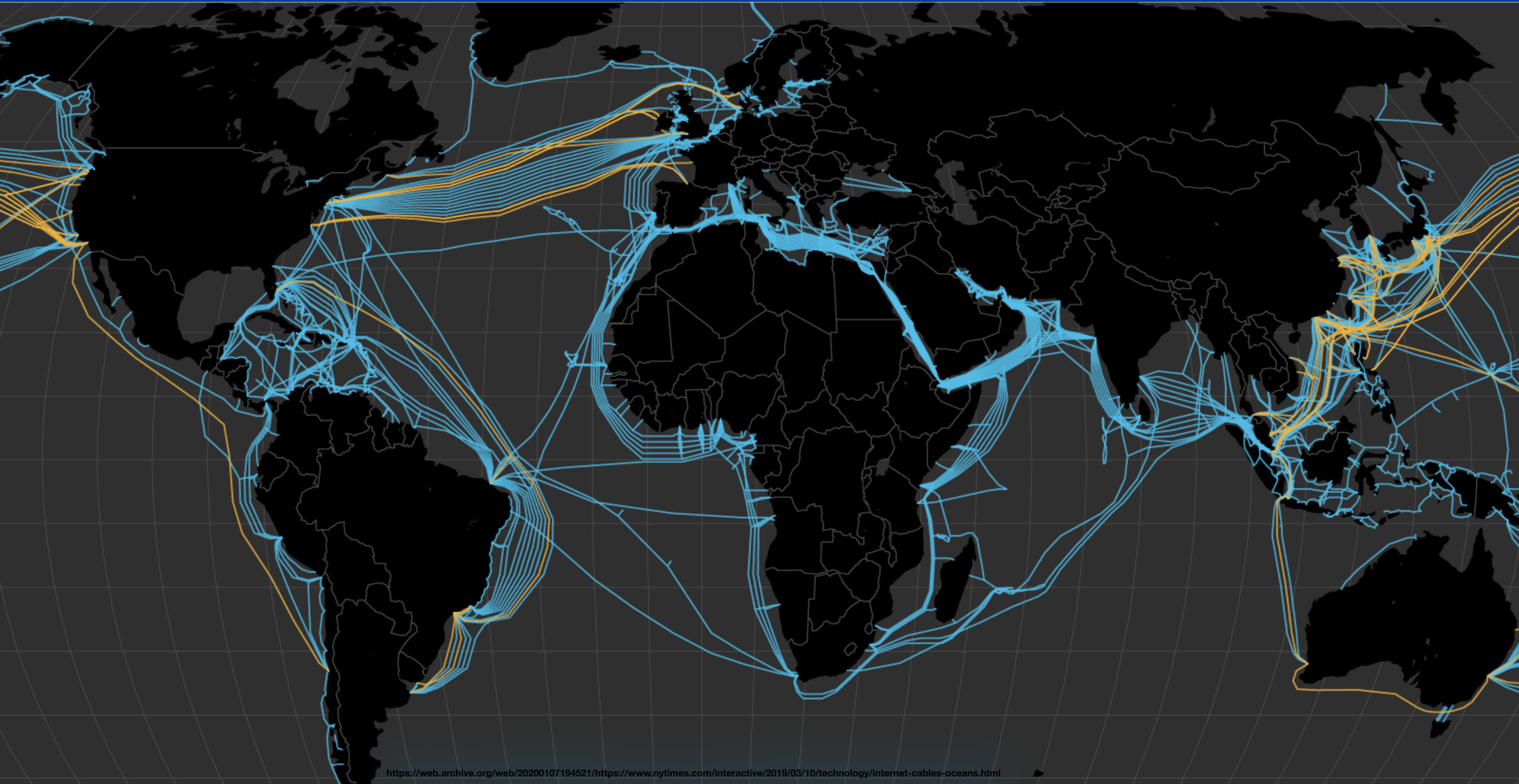
Key Point: Specialized units exploit parallelism or domain-specific hardware for efficiency, complementing the CPU's general role.

Networking Devices

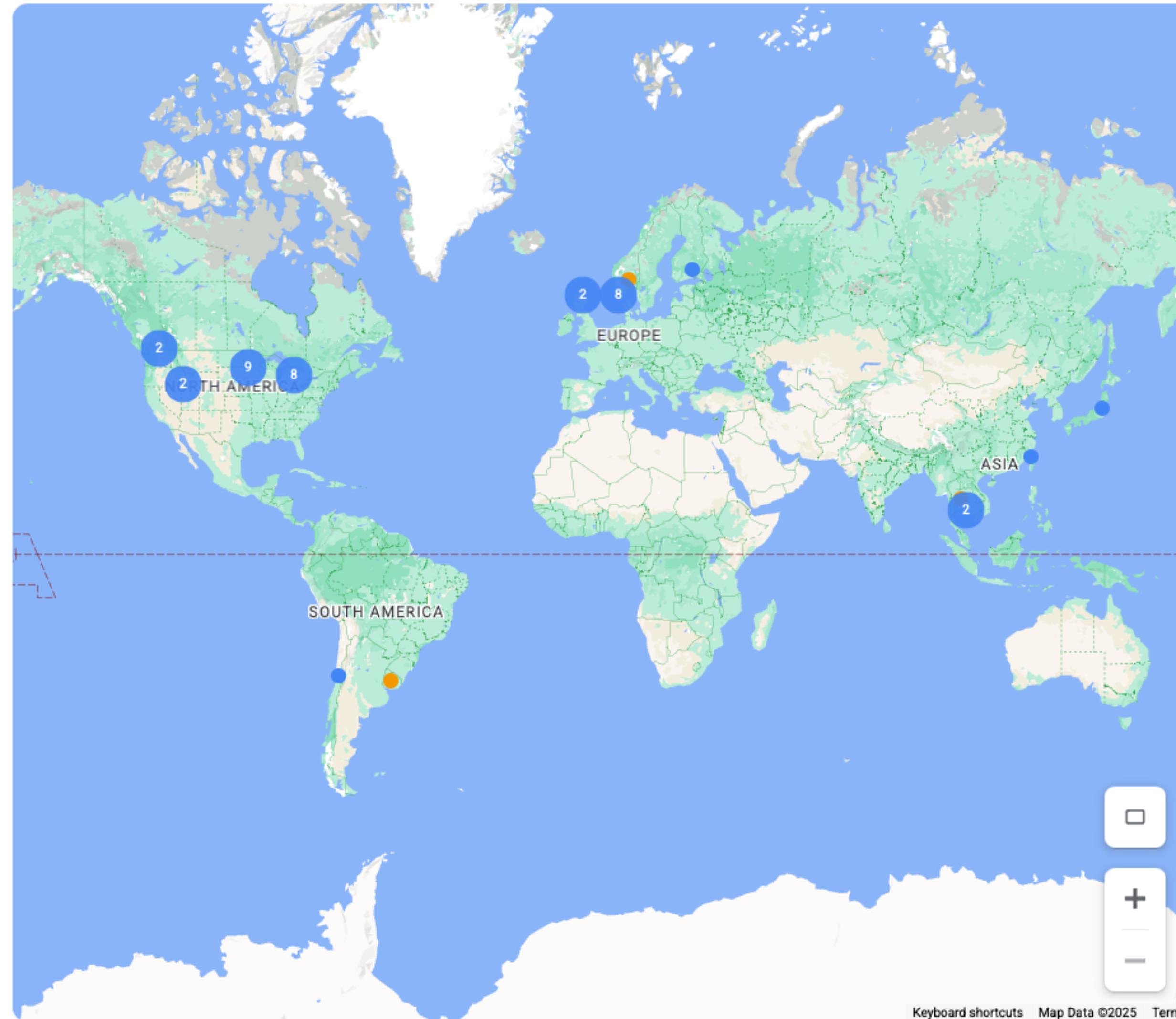
- Connects processing units
 - Within server rack
 - Within data center
 - Across data centers
- Network topologies for efficient distribution of data
 - Spine-leaf topology
 - Fat-tree topology

Low latency

High bandwidth



Google Data Center Locations

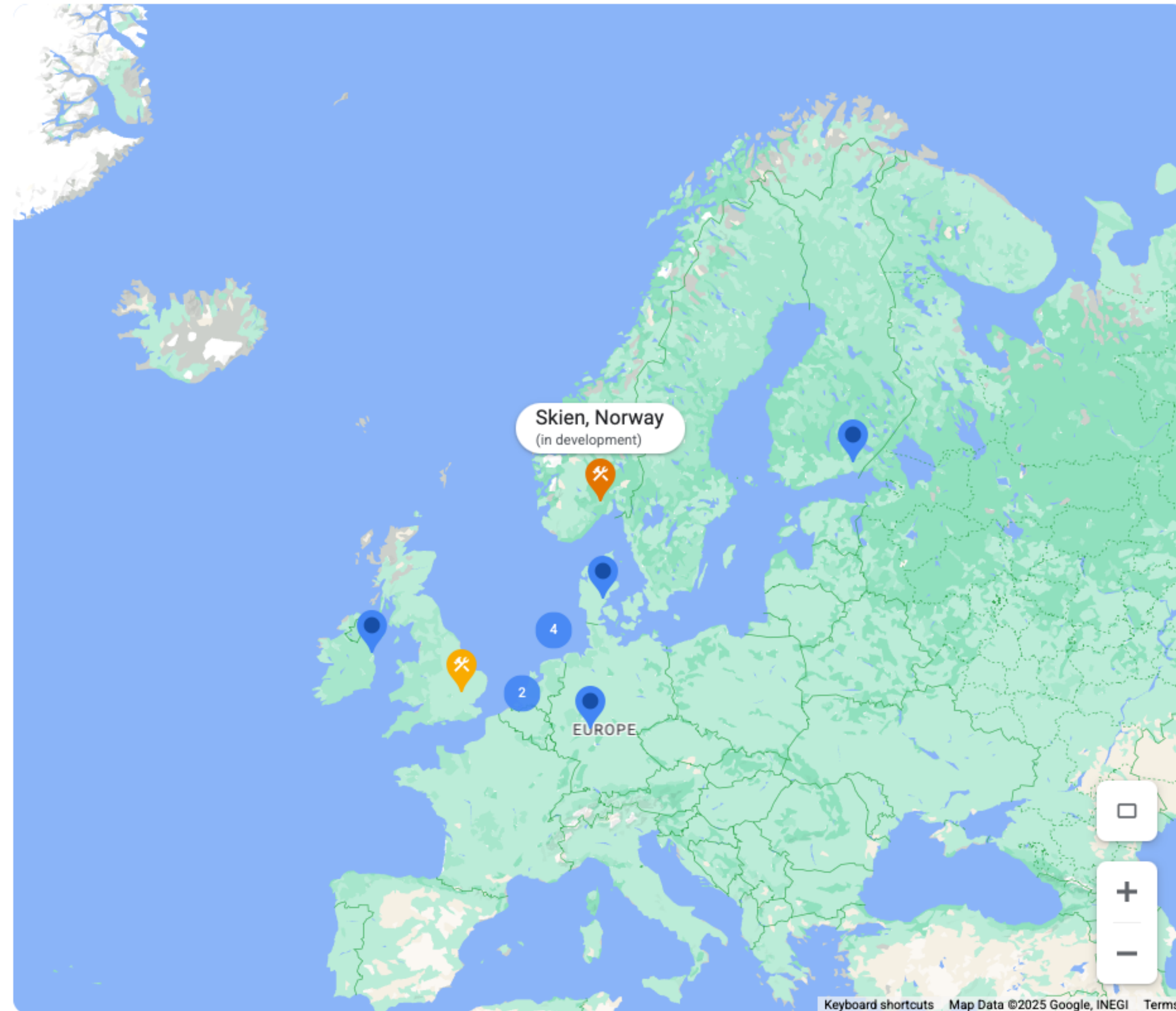


29
locations worldwide

11
countries

Clear Filter

Google Data Center Locations



29
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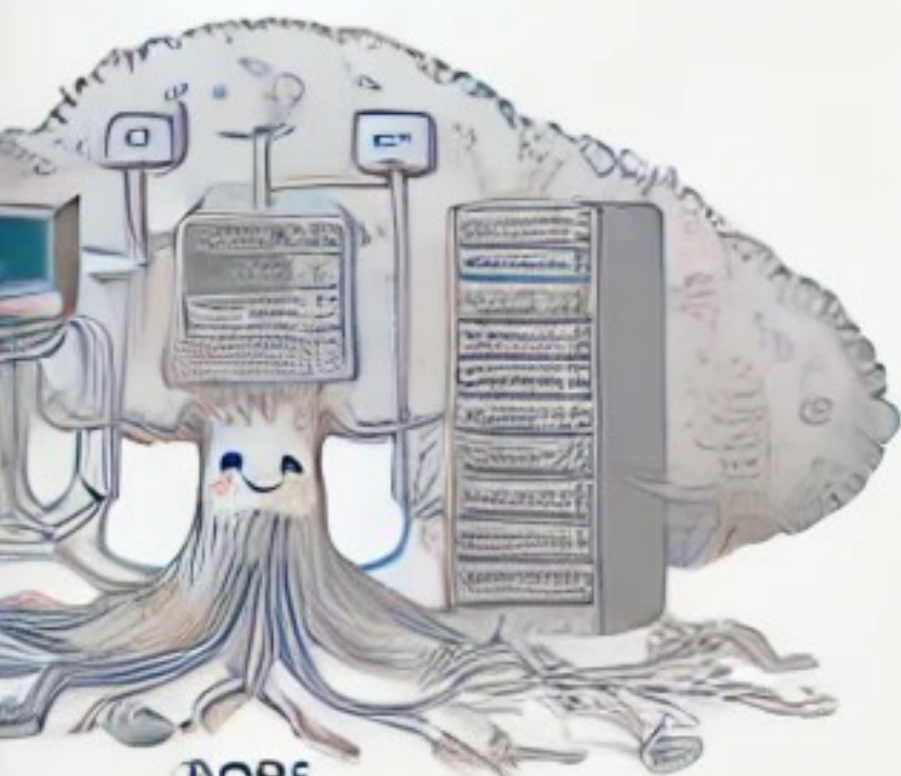
Google Data Center ~ Cloud Computing

How can we connect tens of
thousands of machines in a
data center together?

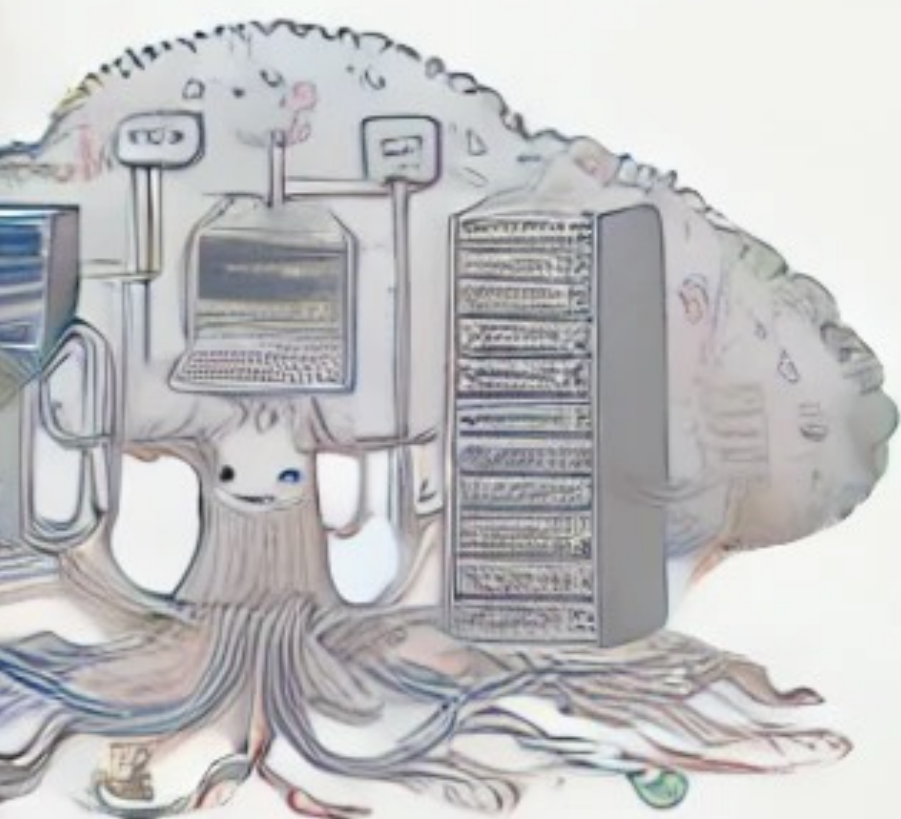




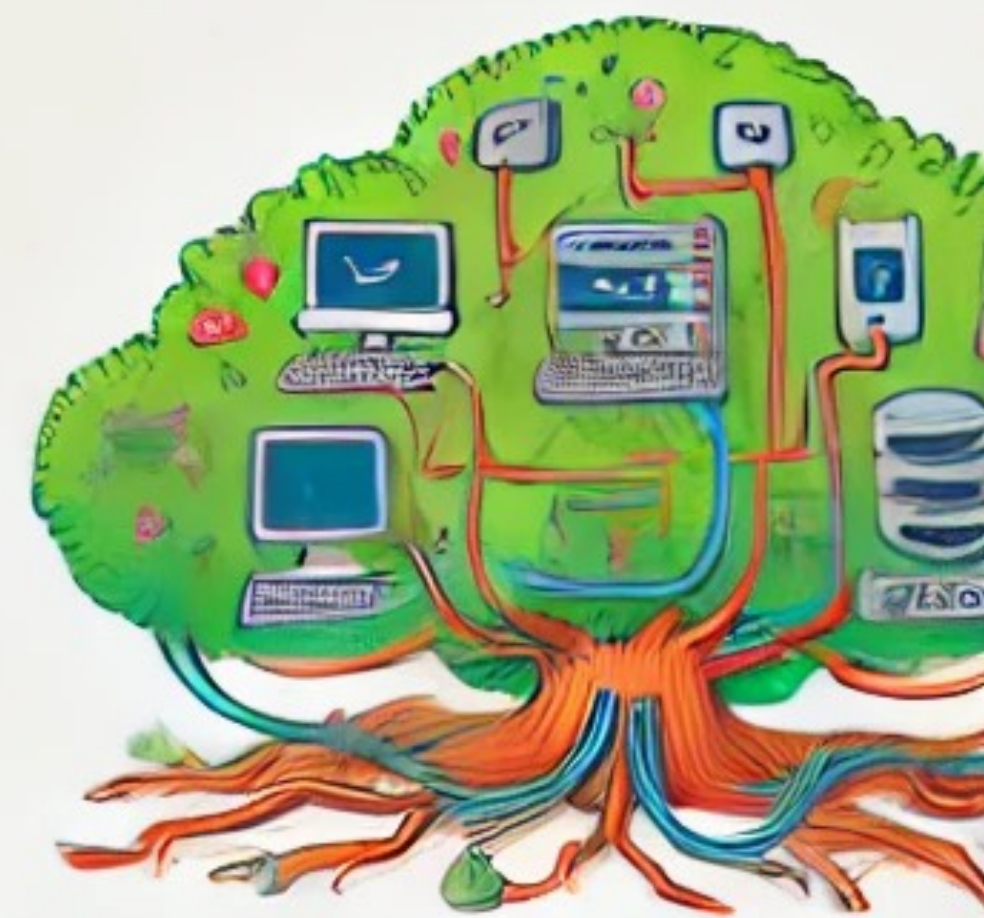
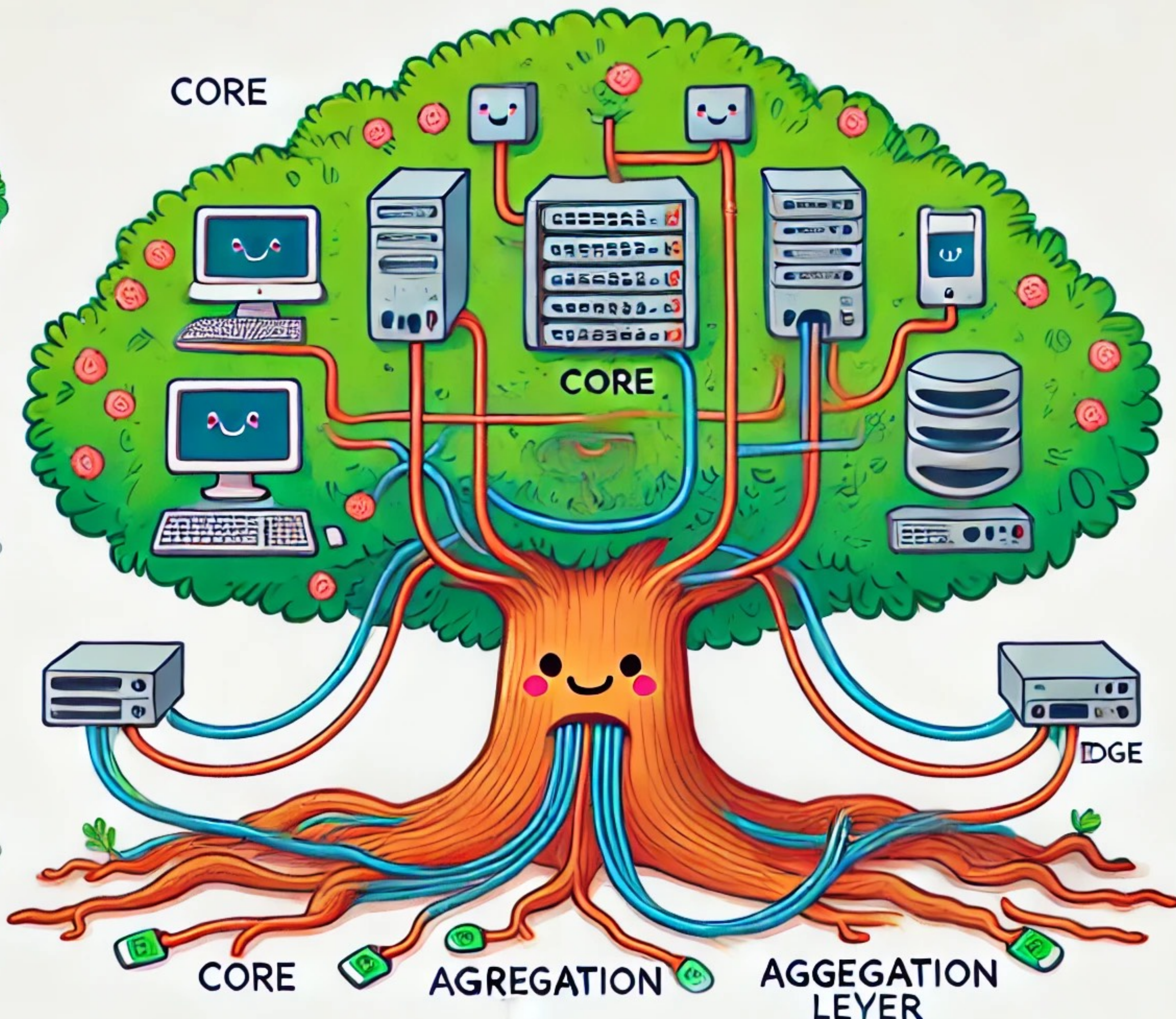
CORE LAYER



CORE



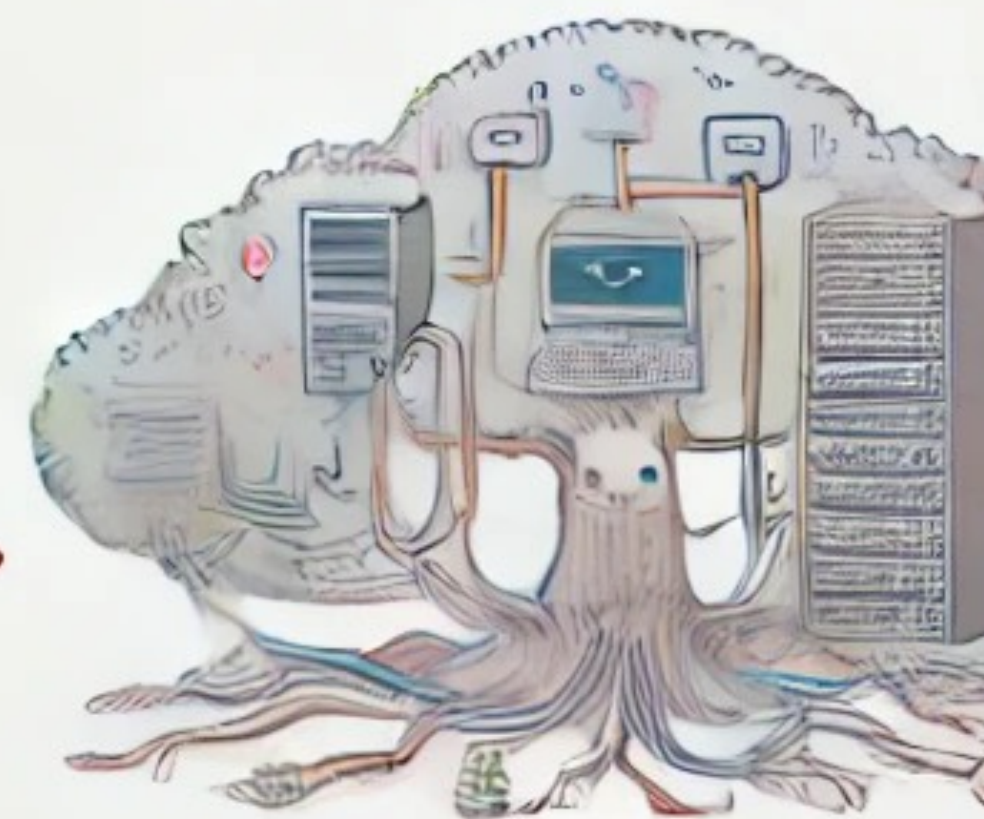
CORE



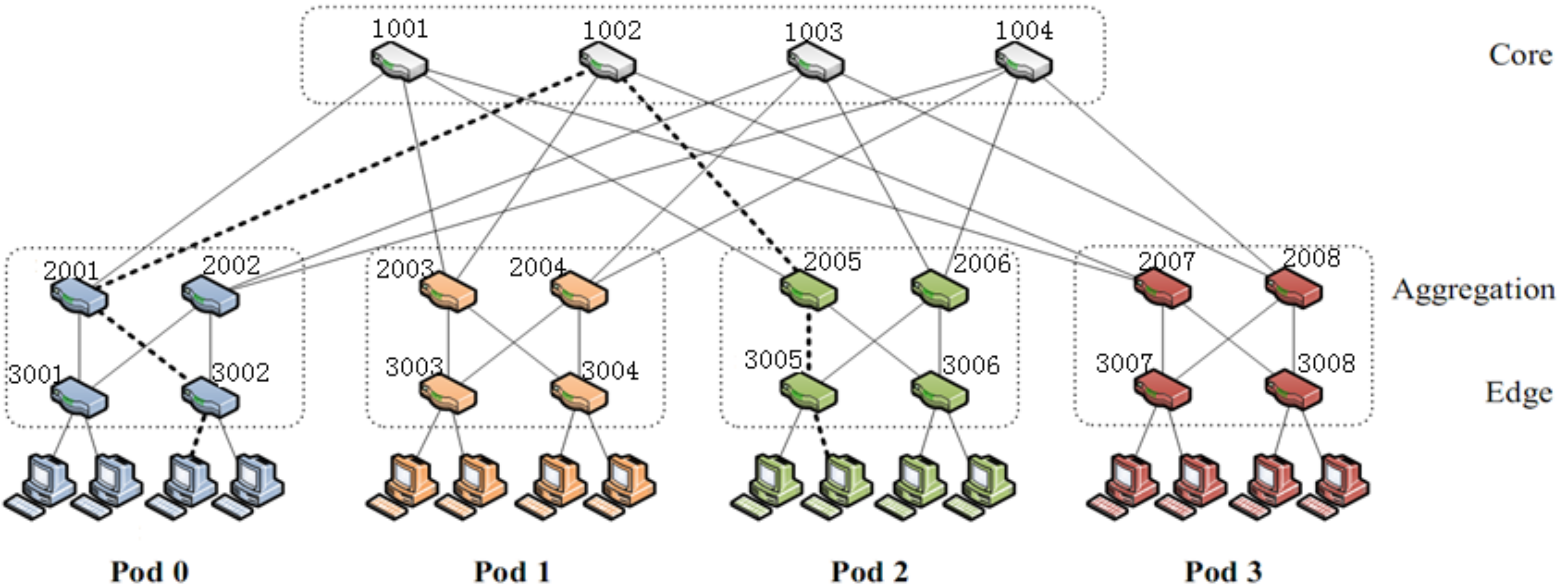
AGREGATION



AGGREGATION



EDGE



Storage Devices Exposed to Cloud Users in Different Ways

- Storage Types
 - Block, Object, and File Storage
- Replication
 - Within single data center
 - Across data centers
 - Across availability zones
- Elastic Storage
 - Adapt storage resources to fluctuating demands
 - Scalability

Virtualization

Recap from Operating Systems

- Virtualizing the CPU
 - The Process Abstraction
- Virtualizing Memory
 - The Address Space Abstraction

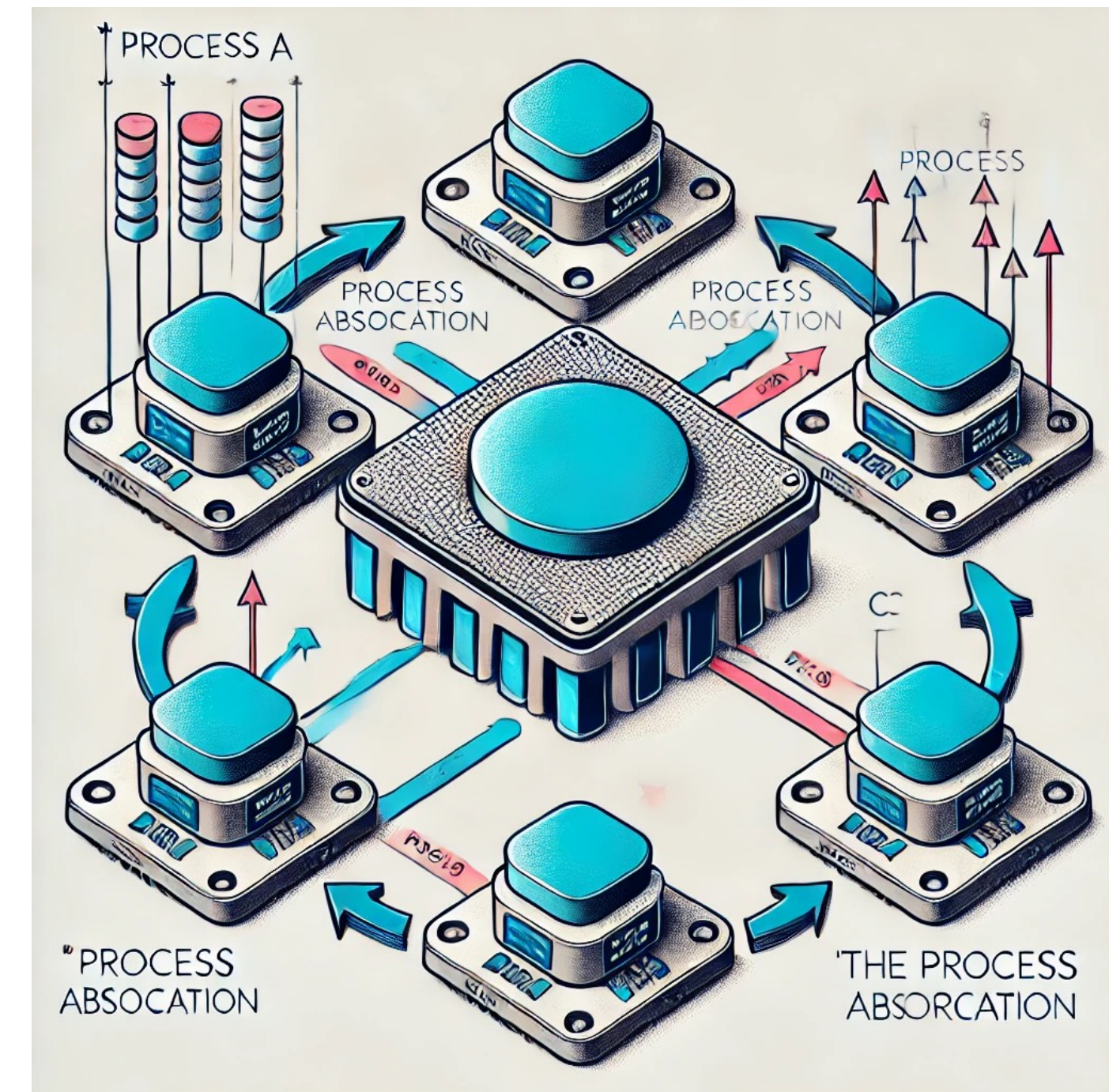


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Why Virtualize?

Resource Sharing

Improve utilization of
(expensive) compute
resources

But Resource Sharing Needs Protection Mechanisms

- **Isolation** between “entities” using a virtual “unit”
 - E.g., two processes running on the same CPU should not interfere with each other

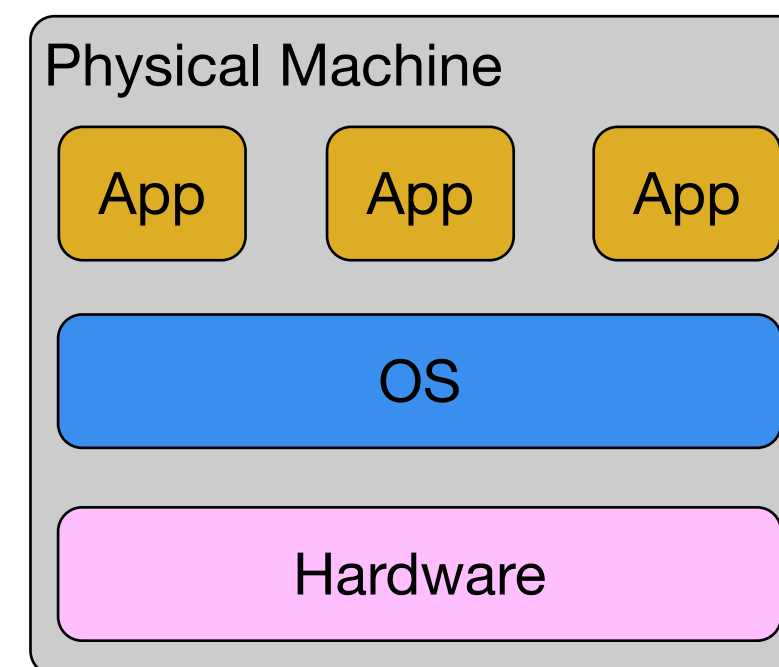
Simplification

Simplification

- **Encapsulation** makes it easy to manage, move, and replicate a virtualized unit
 - Enables scalability and flexibility

The Virtual Machine

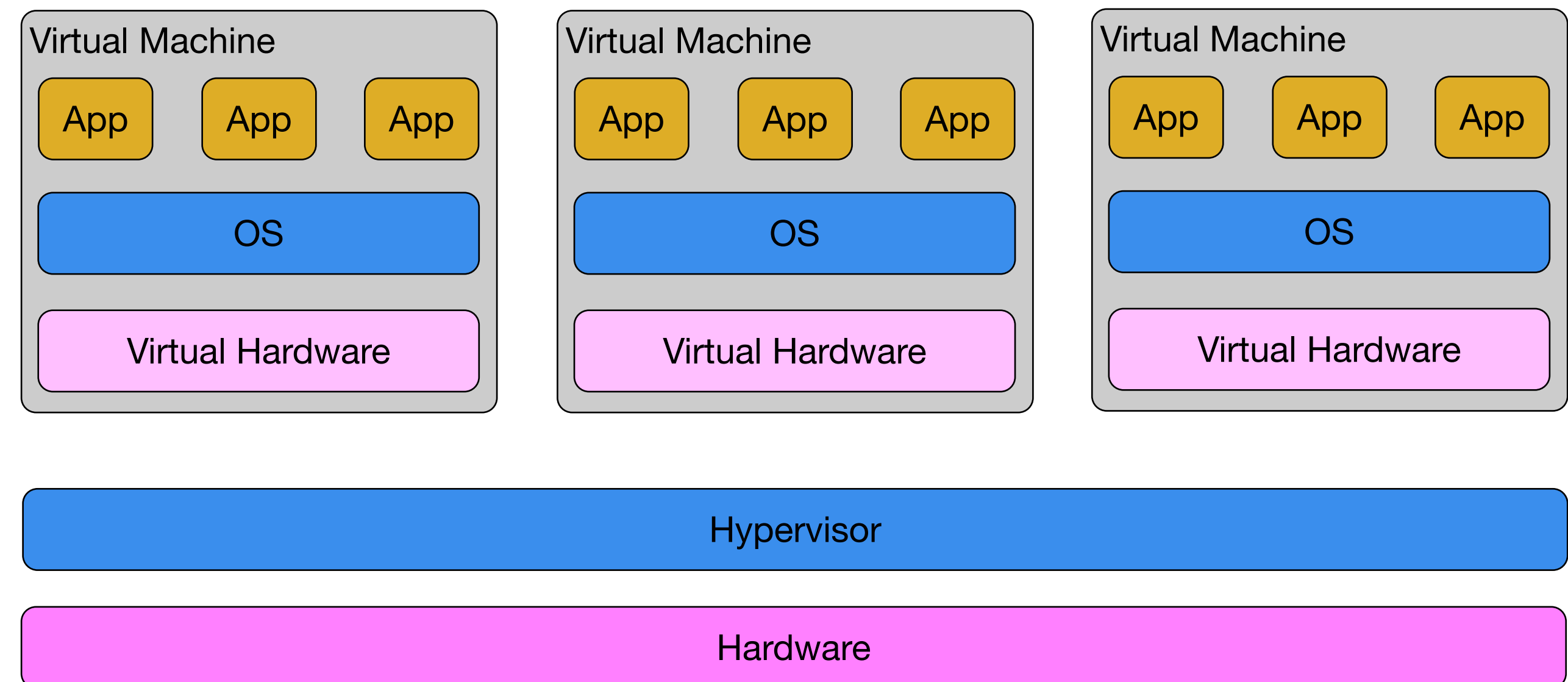
- Physical Machine
 - Processors
 - Memory
 - Storage
 - Network



Bare Metal

The Unit of Abstraction

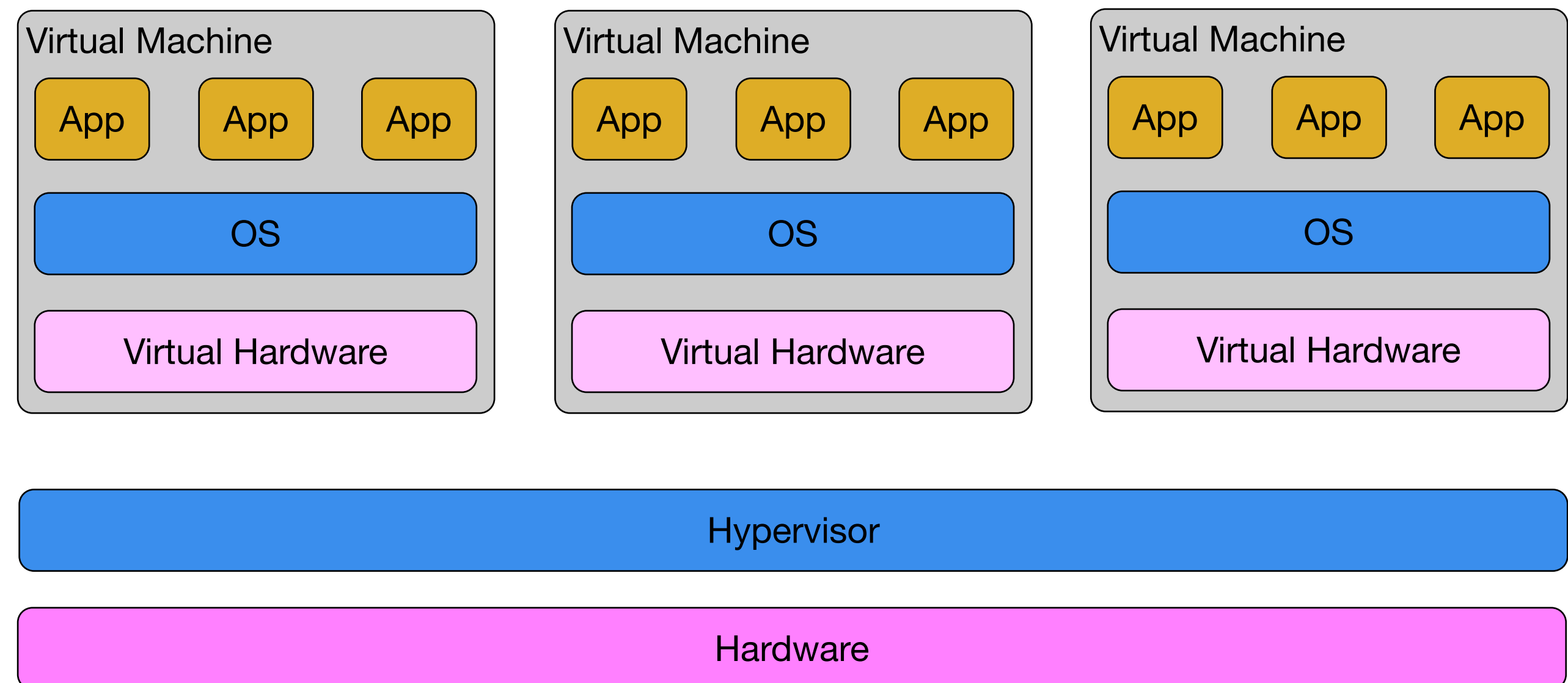
- Virtual Machine — Virtualizing all parts of computer
 - Processors
 - Memory
 - Storage
 - Network



Hypervisor ~
the supervisor above
multiple operating systems

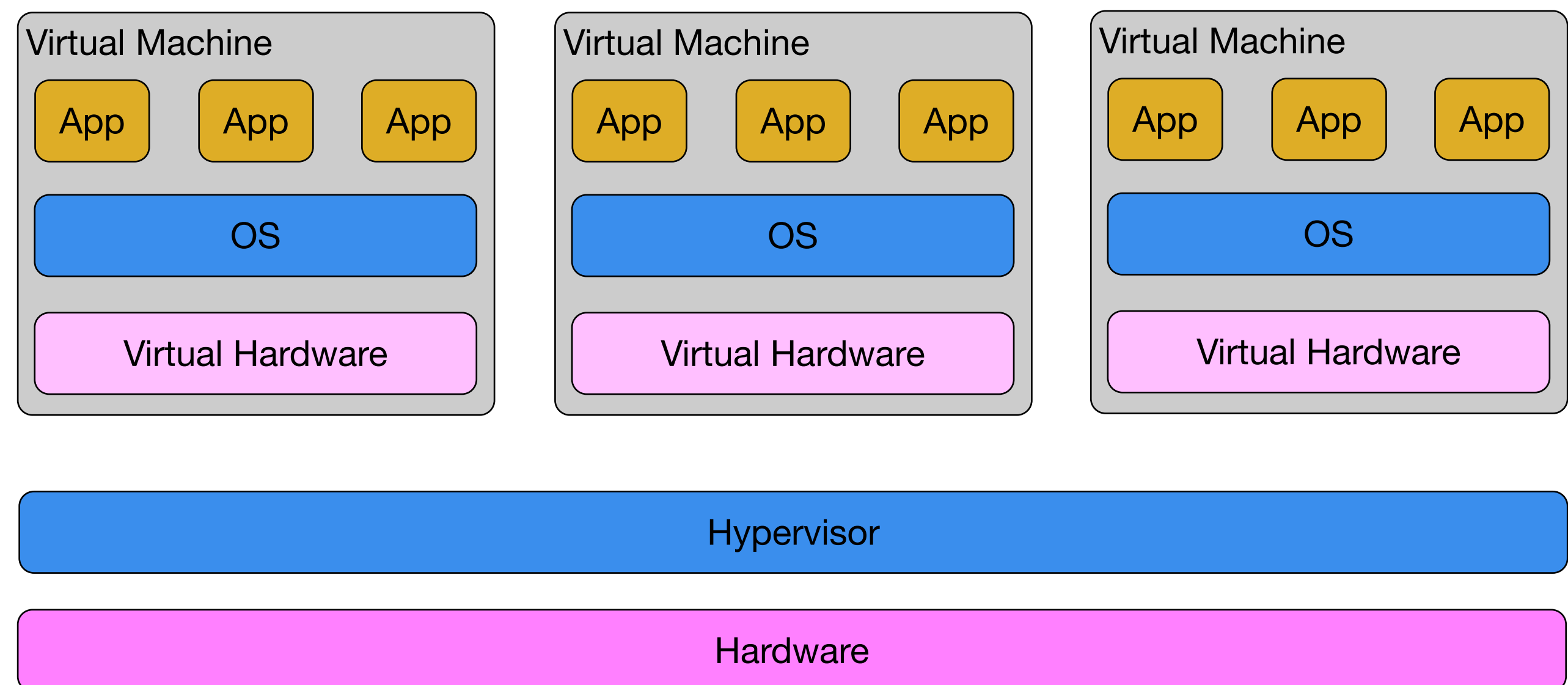
VMs Cannot Interact with Hardware Directly

- Software layer for managing resources and virtual machines
 - Creation, Monitoring, Deletion,
 - Execution of VMs
- Virtual Machine Monitor
= Hypervisor



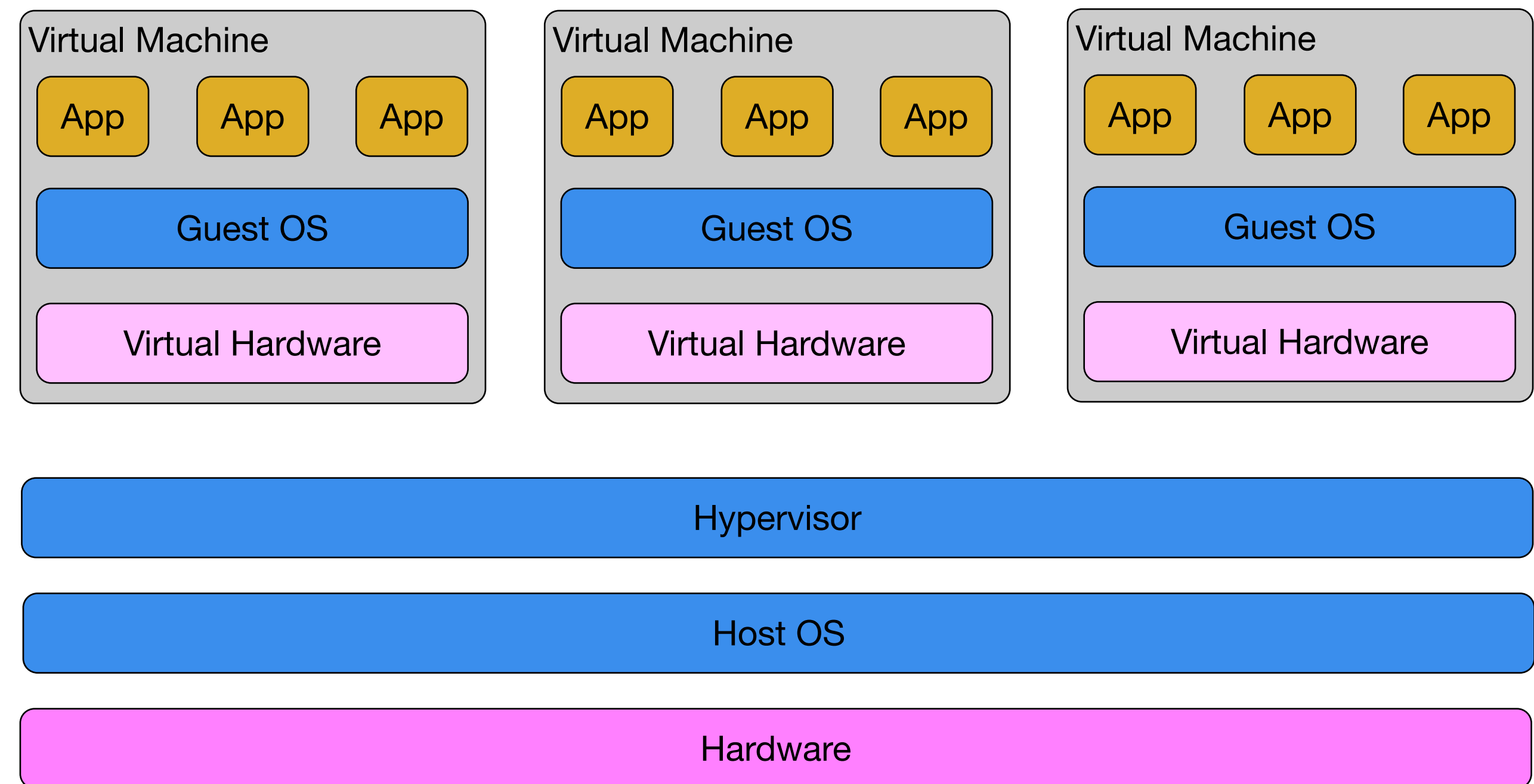
Type 1 Hypervisor (Bare Metal)

- Runs directly on the host's hardware
- Without an underlying OS
- Examples
 - KVM and XEN
 - VMware ESXi and Microsoft Hyper-V



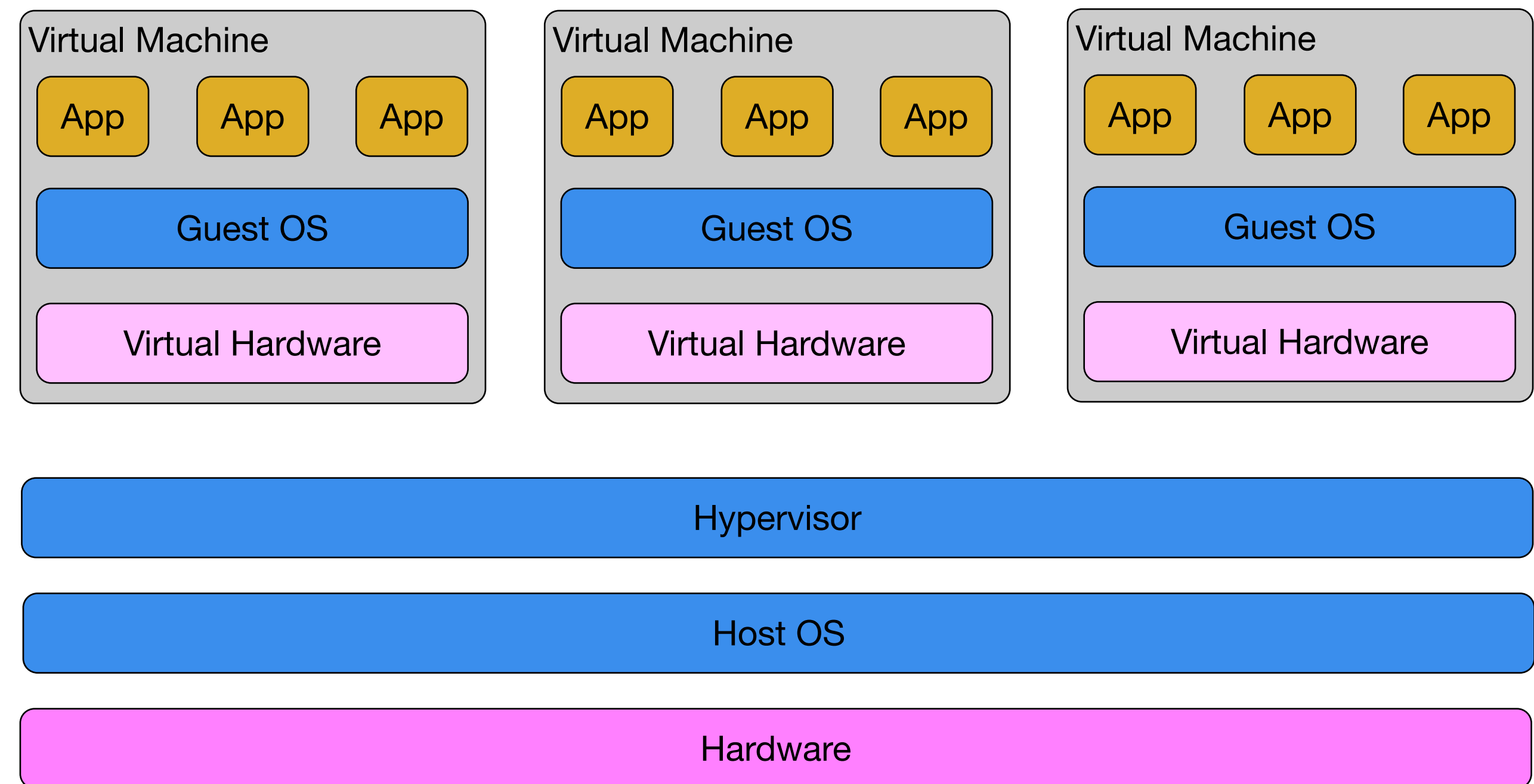
Type 2 Hypervisor (Hosted)

- Runs on top of a host OS
- Relies on the host OS for resource management
- Examples
 - VMware Workstation
 - Oracle VirtualBox



Type 2 Hypervisor (Hosted)

- Desktop app running on host OS
 - Create VMs manually
 - Install guest OS
 - Configure resources via the desktop app
- Typical use case:
 - Run other OS on laptop



Bare Metal vs Virtual Machines

	Bare Metal	Virtual Machines
Runs	Directly on hardware	In virtualized environment
Performance	High (direct access to hardware)	Slightly lower (due to virtualization overhead)
Resource Utilization	Dedicated* (full use of hardware resources)	Shared resources
Isolation	Strong* (physical separation of workloads)	Good* (but relies on hypervisor for isolation)
Flexibility	Low (tied to specific hardware)	High (can run multiple OSes on one host)
Deployment	Slow (physical setup and configuration)	Fast (VMs can be created quickly)
Scalability	Limited to physical hardware capacity	High (easy to provision additional VMs)
Management	Manual	Software-based (hypervisor)
Use Cases	High-performance applications (long-running batch)	General-purpose workloads (with high variability)

Joke time

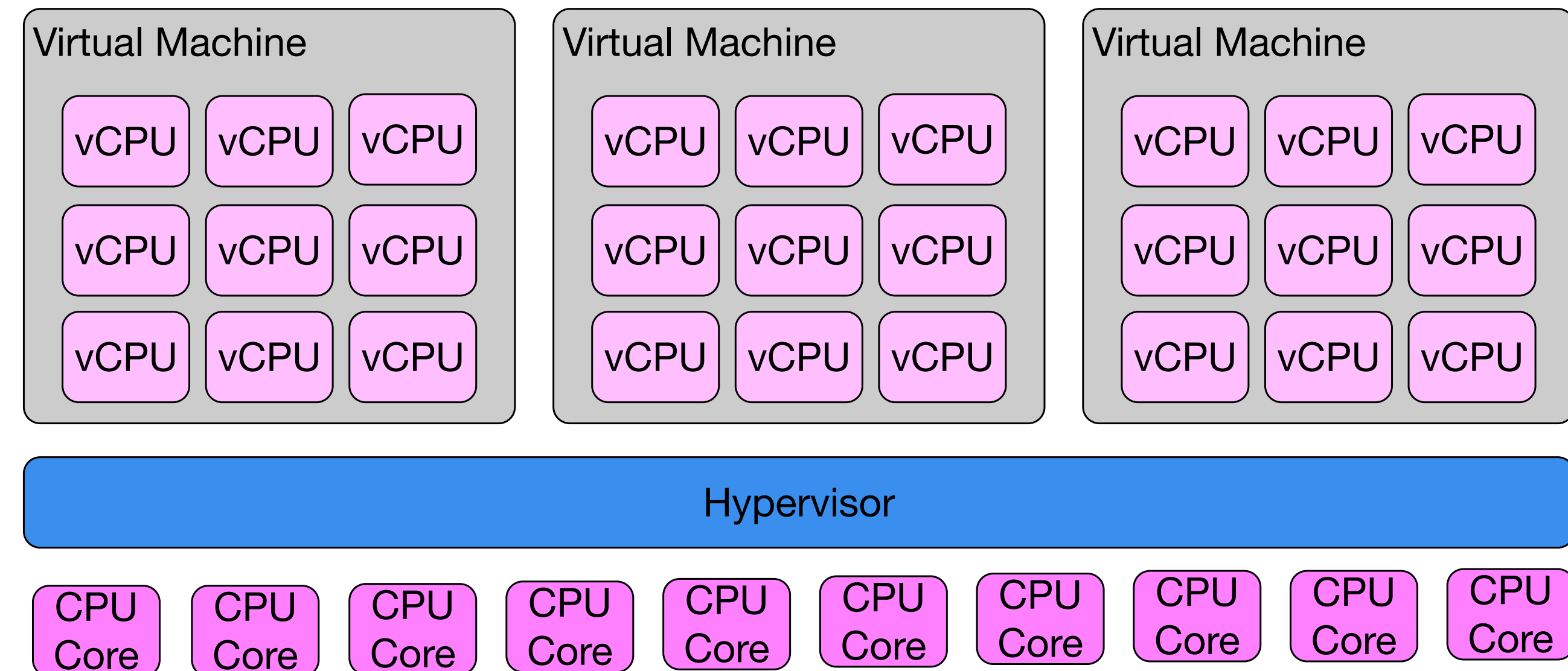
Why did the virtual machine
break up with the hypervisor?

It felt like it was just another
layer of abstraction!

Virtualized Processor Core

Virtual CPU (vCPU) = a portion of the physical CPU resources

- Hypervisor:
 - Divide the physical CPU cores into smaller units called vCPUs
 - Assigns these to virtual machines
 - Schedules vCPUs to run on the physical cores as needed
- vCPU $\neq$ physical CPU core
- A vCPU shares time on a physical core with other vCPUs from different VMs



Virtual CPU (vCPU) = a portion of the physical CPU resources

- The hypervisor manages vCPU allocation to optimize performance
- **Performance:** # of vCPUs assigned to a VM impacts its performance
 - Actual performance also depends on the overall load on the host machine
- **Scalability:** vCPUs allow for flexible and scalable resource allocation
 - Administrators can allocate more/fewer vCPUs to a VM depending on its workload

Virtualizing GPUs



GPU is dedicated to a specific VM or container

- Has direct, exclusive access to the GPU hardware
- **Pros:**
 - Maximum performance, as the workload, has full access to the GPU
- **Cons:**
 - Not resource-efficient for smaller workloads since the GPU cannot be shared

GPU partitioned into multiple smaller instances (hardware level)

- Each instance acts as an independent GPU
 - With dedicated memory and compute cores
- Can be assigned to a different container or VM
- **Pro:**
 - Efficient resource utilization (share a single GPU)
 - Strong isolation between instances prevents instances from affecting each other
- **Cons:**
 - Partitioned instances have limited resources compared to the full GPU

GPU partitioned into multiple virtual GPUs (software level)

- Each instance acts as an independent GPU
 - With dedicated memory and compute cores
- Can be assigned to a different container or VM
- **Pro:**
 - Efficient resource utilization (share a single GPU)
- **Cons:**
 - Partitioned instances have limited resources compared to the full GPU
 - Performance isolation is not as good as hardware-based partitioning

Slurm and Kubernetes

- GPU resources are managed by job schedulers
 - Allocate GPUs to different workloads based on demand and priority
 - Can involve time-sharing or exclusive access, depending on the workload requirements
- **Pro:**
 - Efficient resource management and workload prioritization
- **Cons:**
 - Complexity in scheduling and potential delays due to resource contention

Security

Advantages

- **Isolation:** Compromised VM does not impact other VMs
- **Sandboxing:** For testing potentially unsafe code or applications without risking other parts of the system
- **Snapshots:** Compromised VM can be rolled back using snapshots
- **Access Control:** Hypervisors limit who can interact with VMs and underlying resources
- **Reduced Attack Surface:** Configure VM with minimal amount of services

Disadvantages

- **Hypervisor Vulnerabilities:** Compromised hypervisor exposes all running VMs to compromise
- **Shared Resources:** VMs share hardware; vulnerabilities in resource management (like CPU, memory, or I/O) can potentially be exploited to affect other VMs
 - Rowhammer (DRAM)
 - Side-channel Attacks (CPU): Spectre and Meltdown
- **Complexity:** Complexity of virtualized environments can increase the attack surface, making it harder to secure and manage.

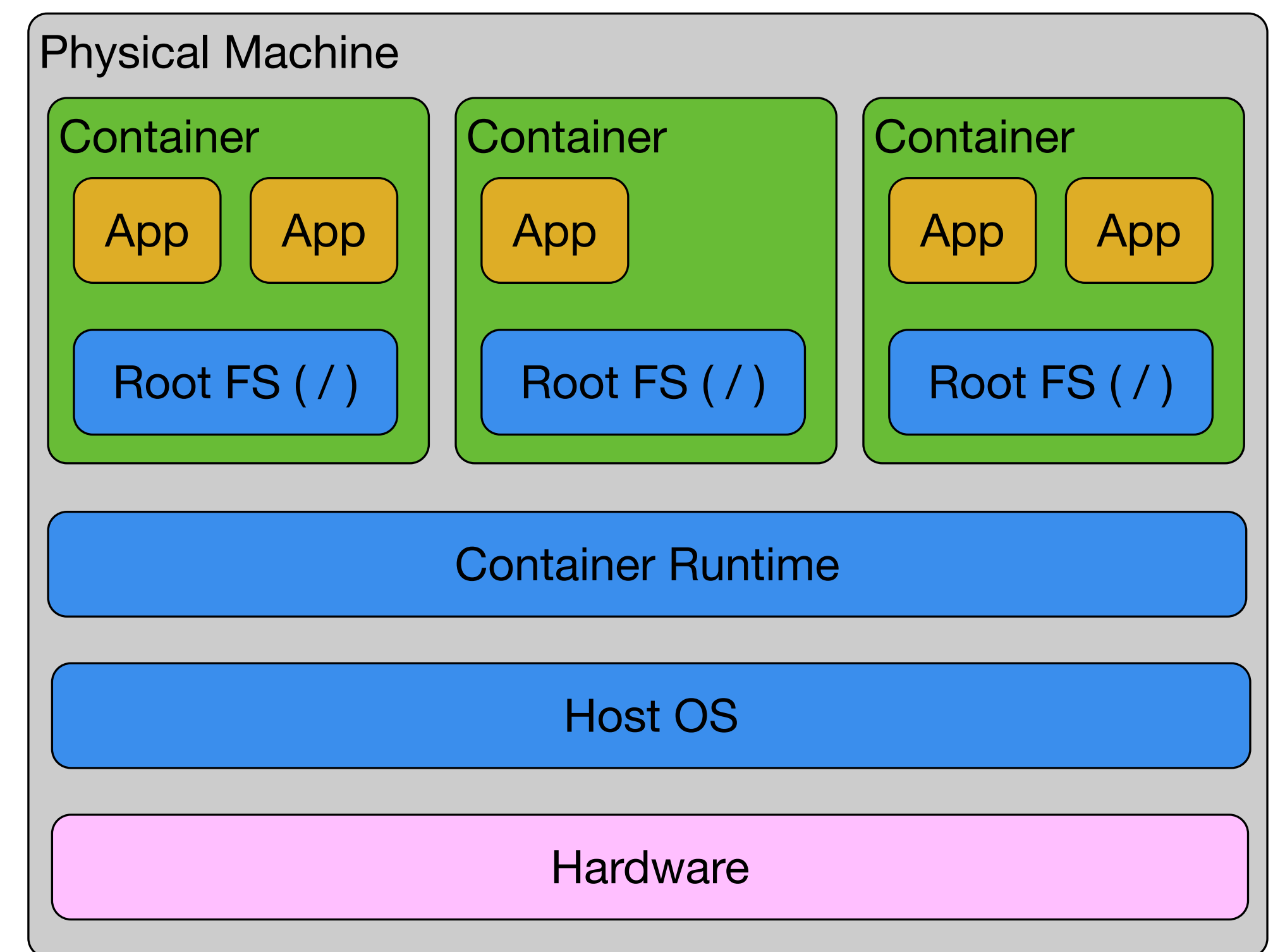
Containers



Lightweight, standalone, and executable software package

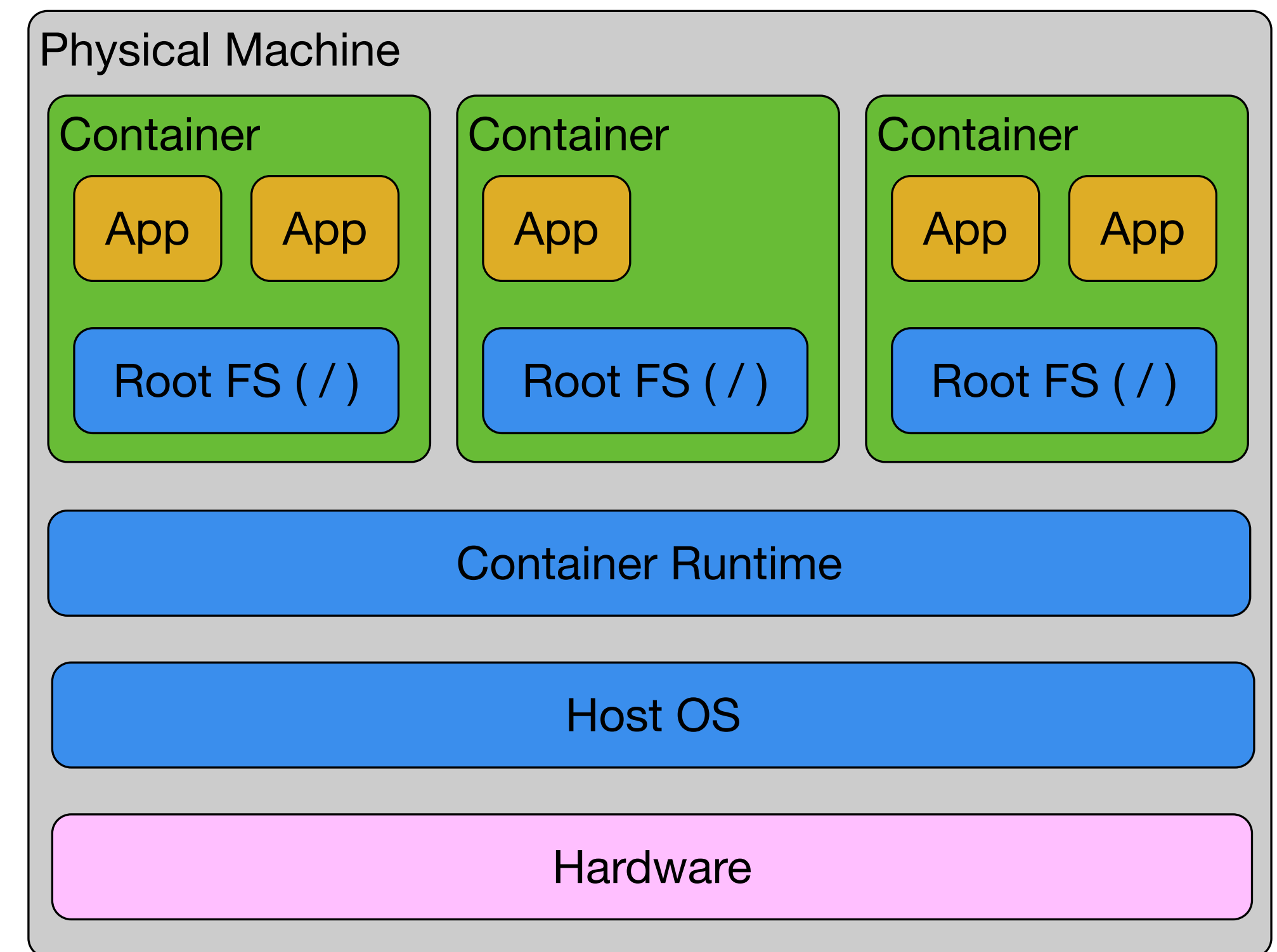
- Application binaries, Configuration files, Runtime libraries, Dependencies
- Pre-configured runtime environment
- Executable with a single command
 - Actually, they are just OS processes

Write once, run anywhere!

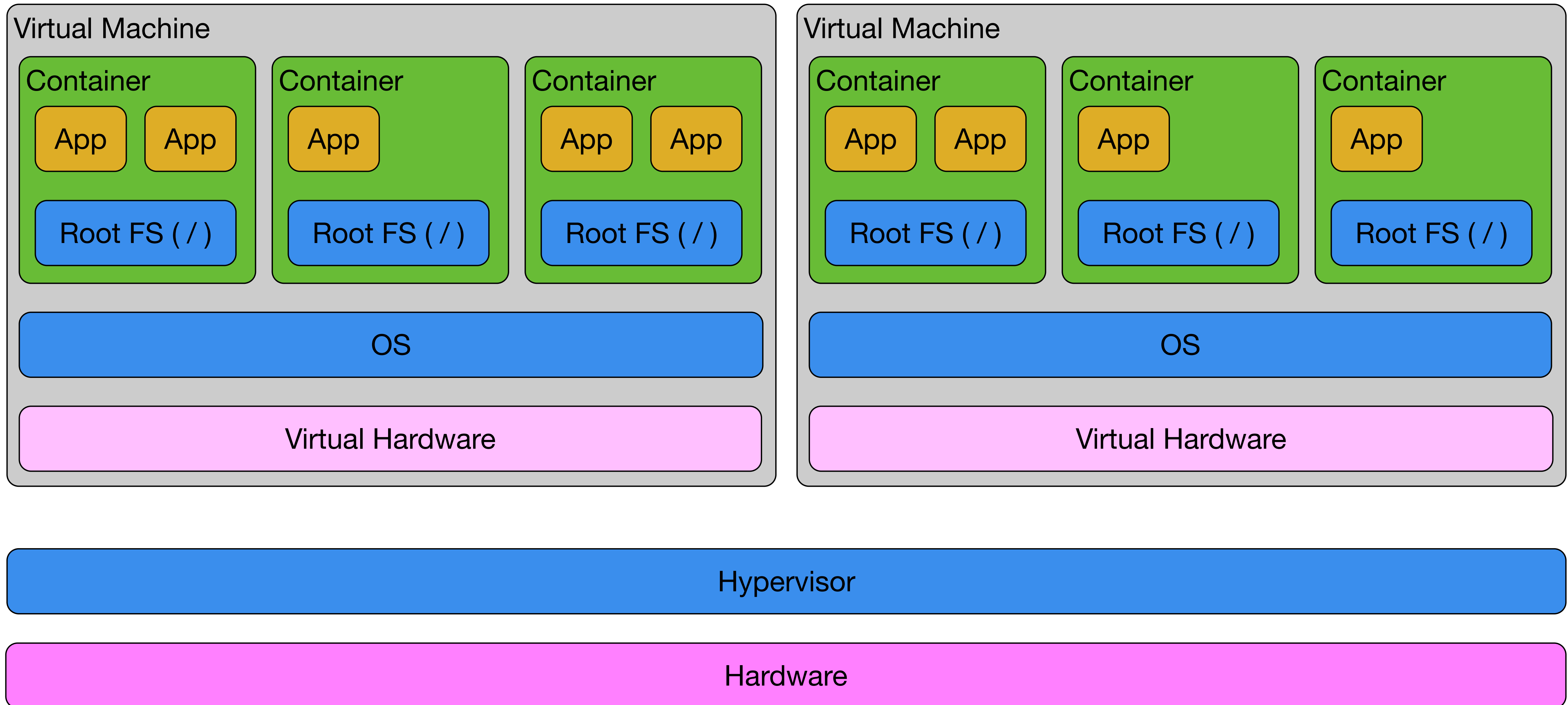


Lightweight, standalone, and executable software package

- Isolate apps from each other
- Standardized packaging format
 - Open Container Initiative (OCI)
- Containers do not include the OS
 - Rely on host OS or container runtime



Containers Often Run on Virtual Machines



Advantages

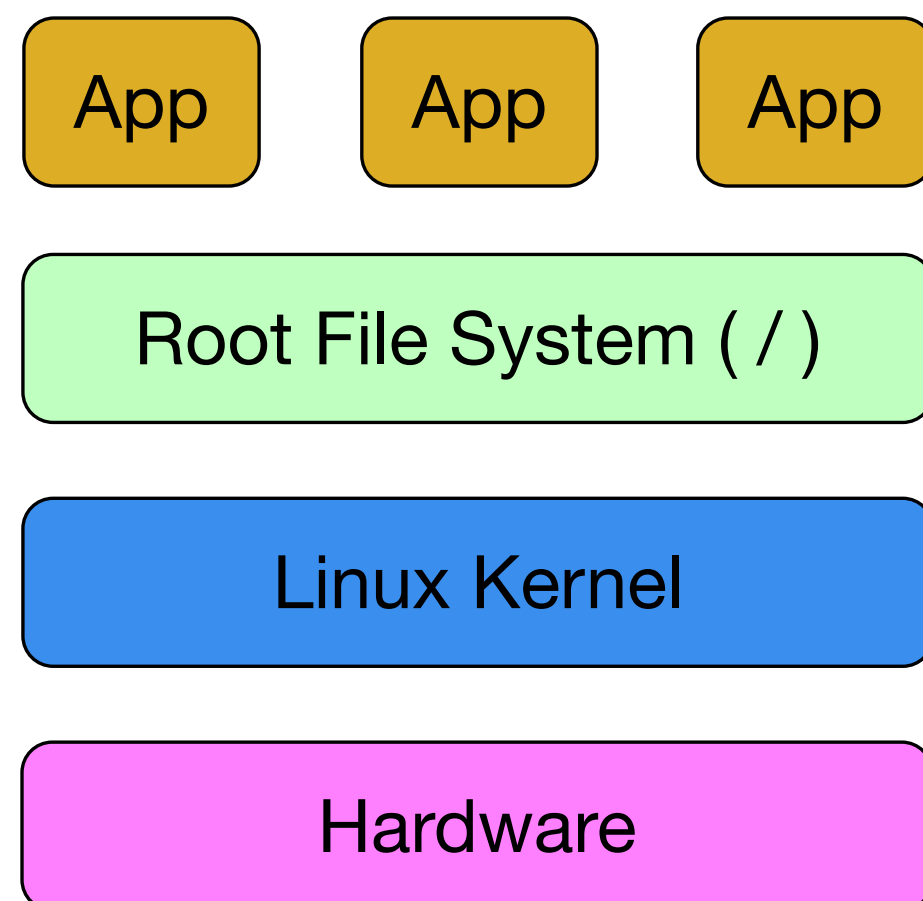
- **Portability:** Run consistently in different environments (development, production)
- **Resource Efficiency:** Share host system's kernel (reduces overhead)
- **Isolation:** Process and filesystem isolation (avoid conflicts between applications)
 - **Security:** Isolating dependencies
- **Rapid Deployment:** Can be quickly started, stopped, and scaled
- **Migration:** Easy to move container between VMs, servers, and cloud machines

Disadvantages

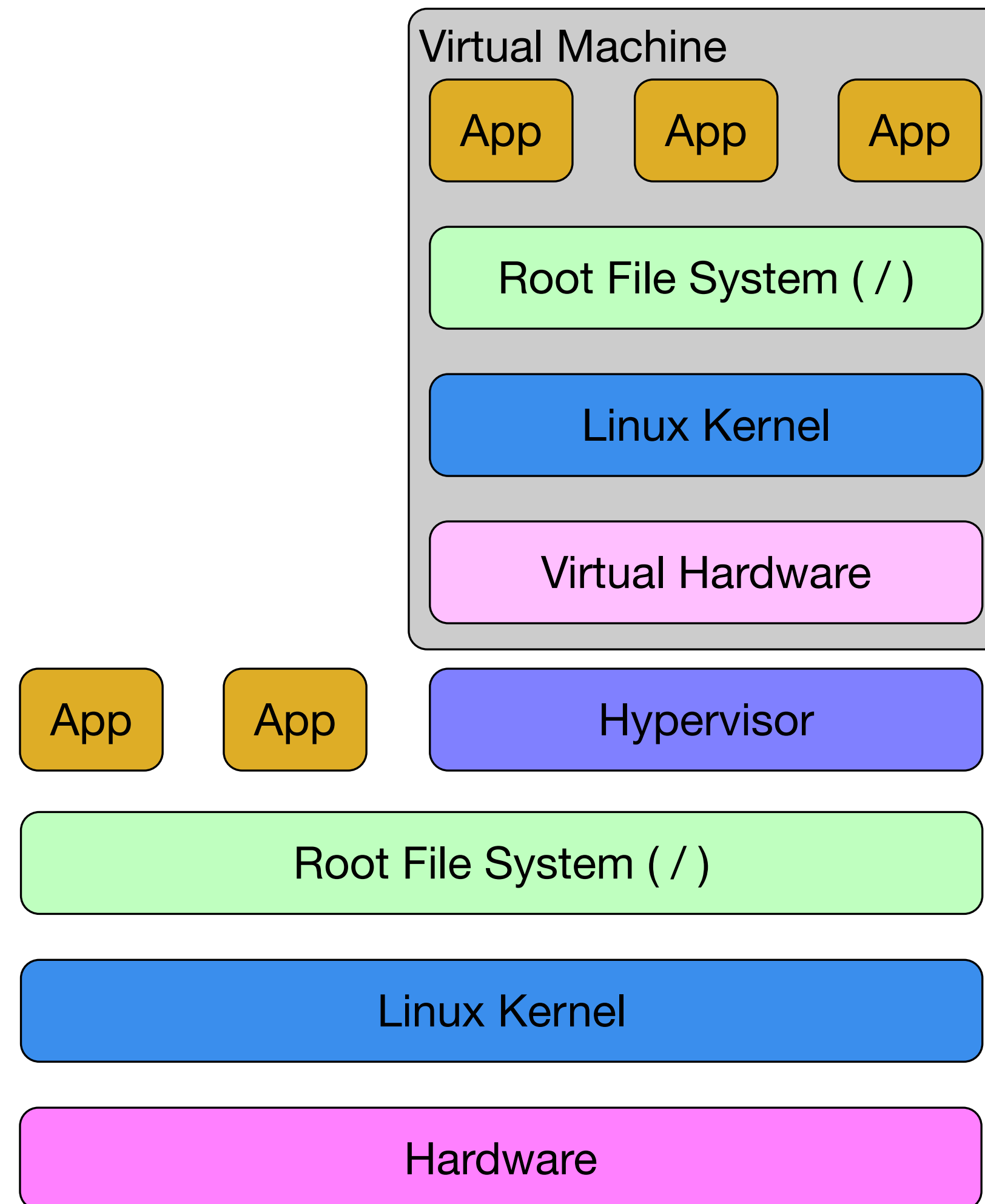
- **Security Concerns:** Compromising the host's kernel can compromise container
- **Networking:** More complex than traditional environments
- **Limited Isolation:** Virtual machines offer stronger isolation than container/
process isolation

Containers vs Virtual Machines

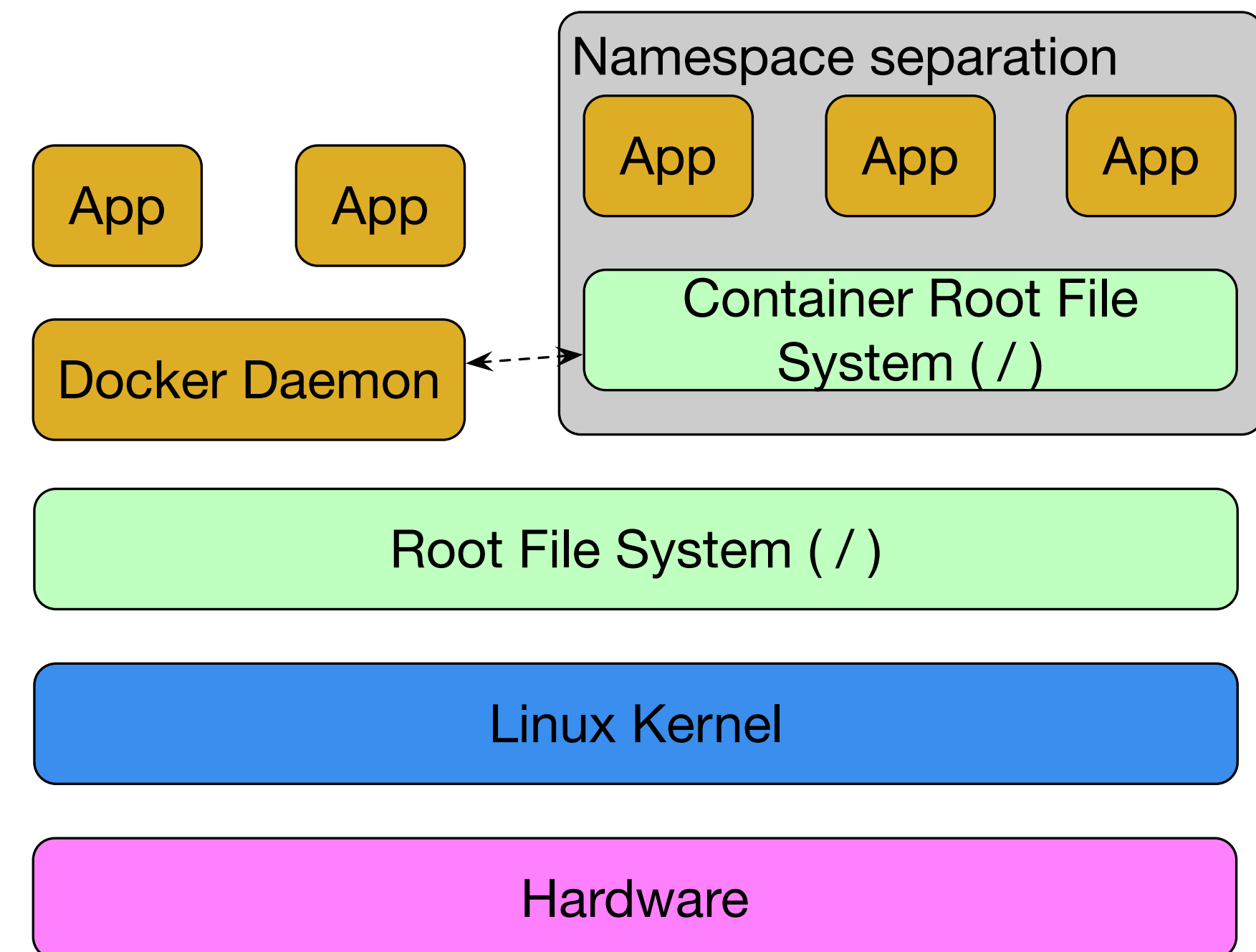
Bare Metal
(No virtualization)

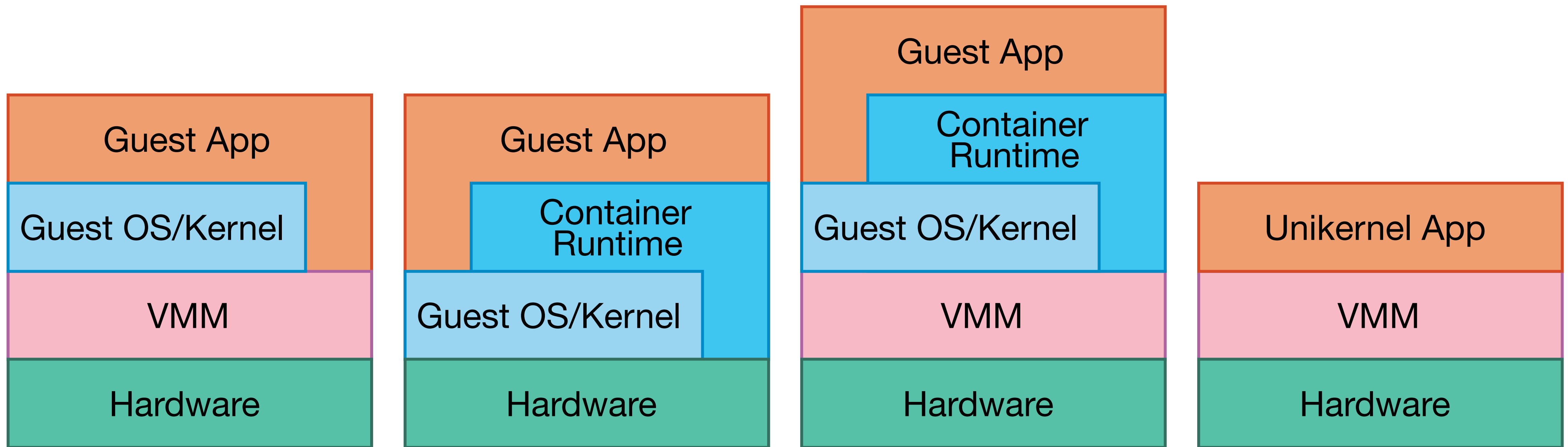


Virtualization



Containers





(a) Traditional Virtual Machines

(b) Containers

(c) Containers in VMs

(d) Unikernels

Specialized OS/kernel

- Lightweight, single-purpose execution environment for applications
- Without any unnecessary components
- Minimize resource usage and attack surfaces
- Unikraft.cloud:
 - Use existing container images without the overhead of actually running containers on Linux

	Containers	Virtual Machines
Boot Time	Seconds	Minutes
Runs on	Host OS Kernel	Hypervisor
Memory Efficiency	High (shares OS kernel)	Low (requires full OS install)
Isolation	Process-level (namespaces)	Strong (full OS isolation)
Deployment	Fast (lightweight)	Slower (heavyweight)
Performance	Near-native (minimal overhead)	Decent, but has some overhead
Scalability	Highly scalable	Less scalable (resource-intensive instances)
Resource Utilization	More efficient	Less efficient
Portability	Highly portable	Less portable
Security	Depends on configuration	Strong due to isolation

History of Virtual Machines and Containers

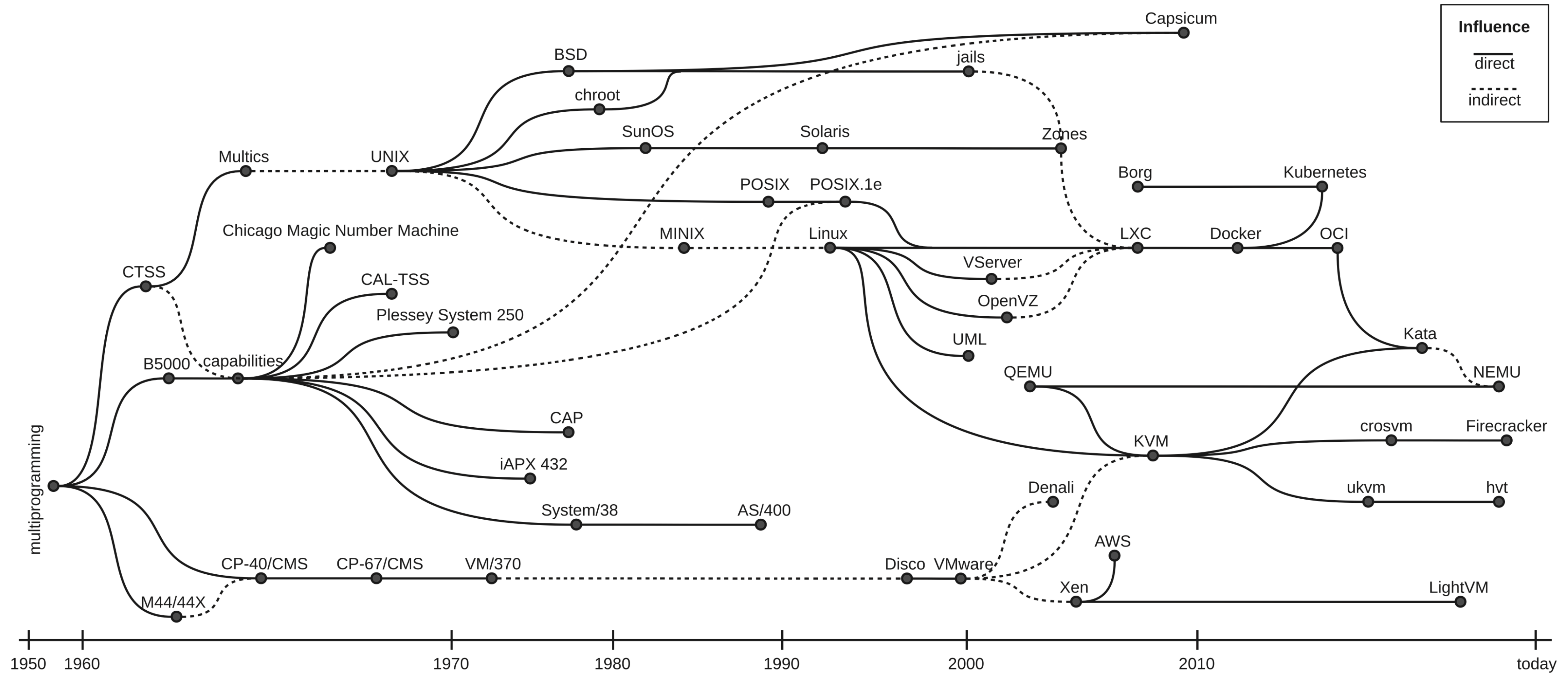


Fig. 1. The evolution of virtual machines and containers.

Source: <https://dl.acm.org/doi/10.1145/3365199>

OCE Containers

- **Image Format Specification:** Defines *standard format* for container images; allows to build, share, and run images on different container runtimes
- **Runtime Specification:** Specifies how a *compliant runtime* should run a OCI-compliant container image
- **Interoperability:** Adhering to OCI standards ensures container technologies from different vendors work together, offering flexibility in choice of tooling

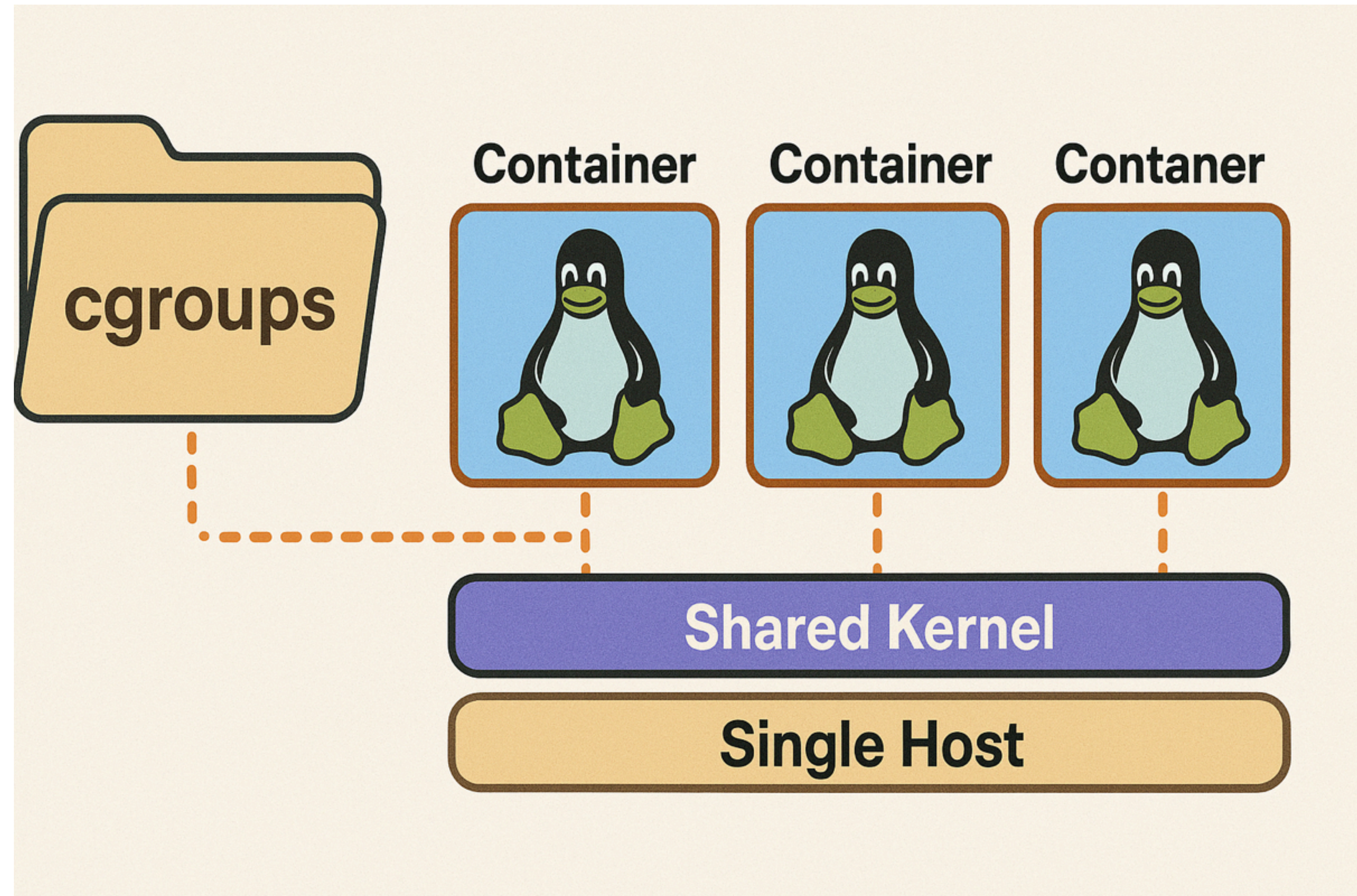


Linux has built-in support for containers

- Allows multiple Linux environments (containers) to run on a single host using a shared kernel
- Provide process isolation via **namespaces** and **control groups**

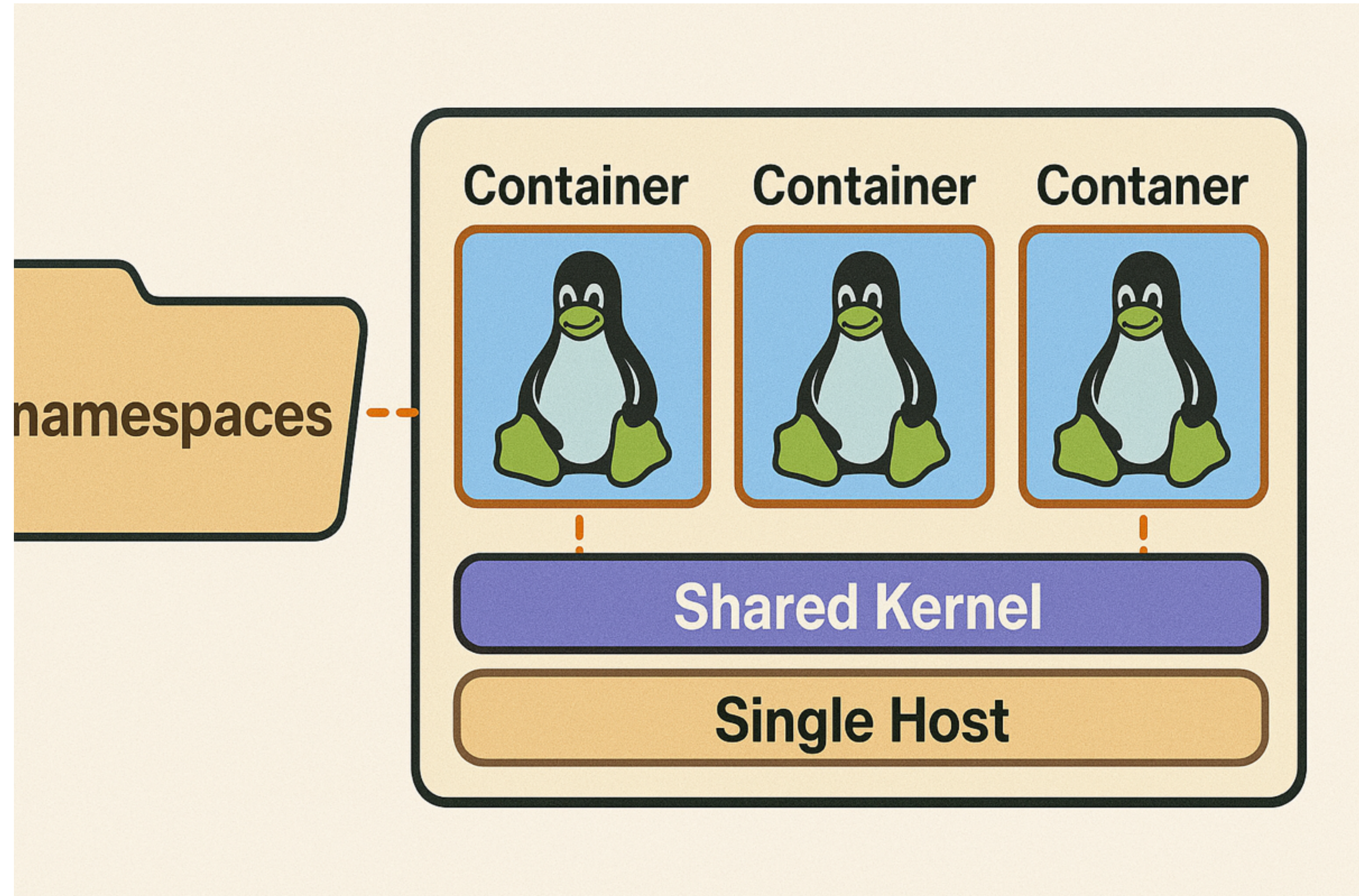
Control Groups

- Controls the resource usage of a collection of processes
 - CPU
 - Memory
 - Disk I/O
 - Network bandwidth



Namespaces

- Create distinct namespaces for different system resources
 - Process ID
 - Network
 - Mount (File system)



Multi-Tenancy

Single-Tenant Architecture

- Each customer (or tenant) has their own dedicated instance of
 - Application
 - Hardware
- Provides full isolation of resources
 - Improved security
 - Better performance
 - Custom hardware (TPUs, Intel SGX, etc.)
 - Customers with stringent security requirements

Multi-Tenant Architecture

- Multiple customers (or tenants) share the same instance of
 - Application
 - Hardware
- Provides logic separation of resources
 - Lower cost
 - Better resource utilization via shared resources (CPU, memory, disk)
 - Shared maintenance
 - Common in SaaS applications (same software platform, different database)

Questions?

- Virtualization — <https://www.ibm.com/topics/virtualization>
- Virtual Machines — <https://www.ibm.com/topics/virtual-machines>
- Hypervisors — <https://www.ibm.com/topics/hypervisors>
- Containers — <https://www.ibm.com/topics/containers>
- Unikernels — <https://unikraft.cloud/>